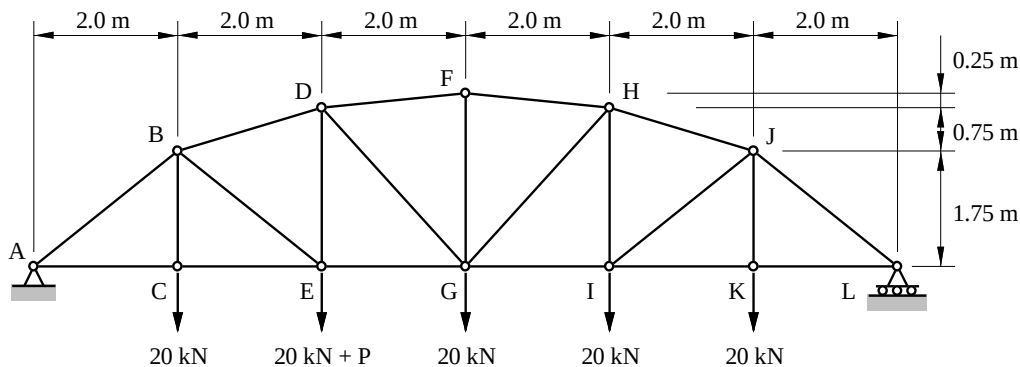


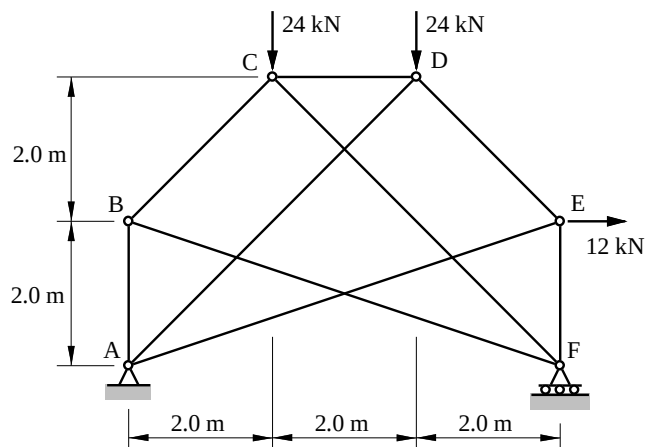
UNIVERSITY OF BIRMINGHAM
SCHOOL OF ENGINEERING
Mechanics-1: Tutorial 18– Truss Structures-2

4. In the pin-jointed bridge truss shown in the diagram, the dead load of the deck acts as a vertical force of 20 kN at each of the joints of the lower chord. Inspection has revealed a defect in member DG - and, so that it can be removed and replaced, the force in it is to be reduced to zero by applying an additional vertical load (P) at joint E.



Use the Method of Sections to find the total axial force in DG, in terms of P. What is the value of P that should be applied? [Ans: 4.29 kN]

5. The diagram shows a pin-jointed frame with nine members, three reaction components and six joints (members cross without touching where no pins are shown). Is the frame statically determinate? Can it be analysed by the Method of Joints?



Calculate the external reactions and use the Method of Sections to find the internal forces in members AB, CD and EF.

Ans: $V_A = 20 \text{ kN}$, $H_A = 12 \text{ kN}$, $V_F = 28 \text{ kN}$; $T_{AB} = -16 \text{ kN}$, $T_{CD} = 0$, $T_{EF} = -20 \text{ kN}$

UNIVERSITY OF BIRMINGHAM
SCHOOL OF ENGINEERING
Mechanics-1: Solutions Tutorial 21 & 22– Truss Structures

1.	Question	m	r	j	m+r	2j	Comment
a		12	4	8	16	16	Statically determinate
b		10	3	7	13	14	Mechanism
c		13	4	8	17	16	Statically indeterminate
d		5	4	4	9	8	Statically indeterminate
e		11	3	7	14	14	Statically determinate

2a Resolve horizontally:

$$H_E + H_F = 0, \Rightarrow H_E = -H_F$$

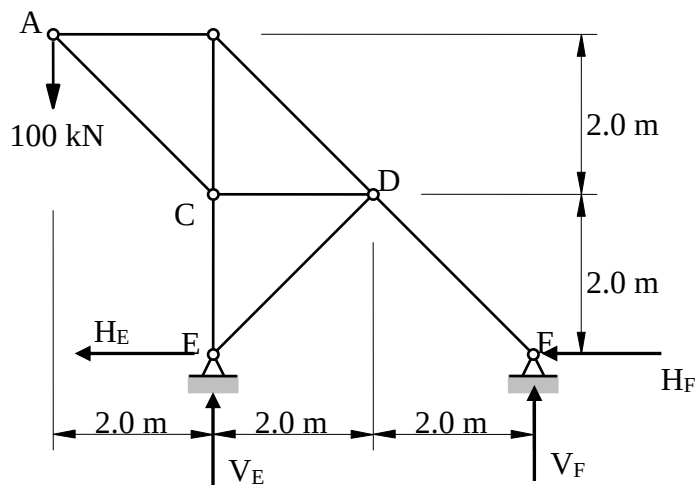
Resolve vertically:

$$V_E + V_F - 100 = 0 \Rightarrow V_E + V_F = 100$$

Take moments about E (+ve clockwise)

$$-(100 \times 2) - (4V_F) = 0 \Rightarrow V_F = -50 \text{ kN}$$

$$\therefore V_E = 150 \text{ kN}$$



Consider Joint F

Resolve vertically: $T_{DF} \cos 45^\circ + V_F = 0 \Rightarrow T_{DF} = -(-50)\sqrt{2} = 70.7 \text{ kN}$

Resolve horizontally $T_{DF} \cos 45^\circ + H_F = 0 \Rightarrow H_F = -50 \text{ kN}$

Using method of joints, move through joints in the order A, B, C and then D calculating the member forces. Check your answers at joint E.

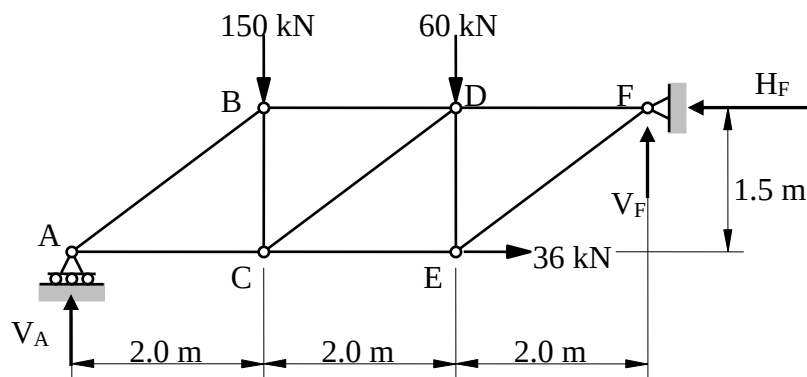
2b Resolve vertically:

$$V_A + V_F - 150 - 60 = 0$$

$$V_A + V_F = 210$$

Resolve horizontally:

$$H_F = 36 \text{ kN}$$



Taking moments about F (+ve clockwise)

$$(V_A \times 6) - (150 \times 4) - (60 \times 2) - (36 \times 1.5) = 0 \Rightarrow V_A = 129 \text{ kN, hence}$$

$$V_F = 81 \text{ kN}$$

Using the method of joints, move through the joints in the order A, B, C, E and D calculating the member forces. Check at joint F.

3 Resolve vertically: $V_A + V_H - 70 - 10 = 0$. $V_A + V_H = 80$

Resolve horizontally: $H_A = 0$

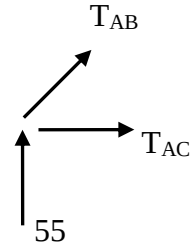
Take moments about point A (+ve anticlockwise)

$(70 \times 1) + (10 \times 3) - (4V_H) = 0$. $\Rightarrow V_H = 25 \text{ kN}$, hence $V_A = 55 \text{ kN}$.

Consider joint A

Resolve vertically: $T_{AB}\cos 45 + 55 = 0$
 $\therefore T_{AB} = -77.8 \text{ kN}$ (compression)

Resolve horizontally: $T_{AB}\cos 45 + T_{AC} = 0$
 $\therefore T_{AC} = 55 \text{ kN}$ (tension)

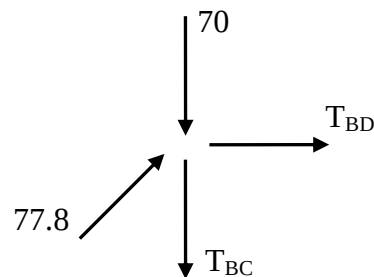


Assumption 1: CD is in tension. Hence, no force exists in BE and this member can be ignored.

Consider joint B

Resolve vertically: $77.8\cos 45 - 70 - T_{BC} = 0$
 $\therefore T_{BC} = -15.0 \text{ kN}$ (compression)

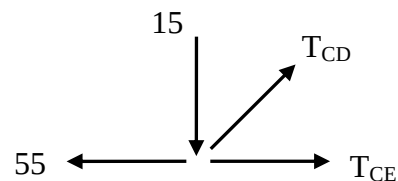
Resolve horizontally: $77.8\cos 45 + T_{BD} = 0$
 $\therefore T_{BD} = -55 \text{ kN}$ (compression)



Consider joint C

Resolve vertically: $T_{CD}\cos 45 - 15.0 = 0$
 $\therefore T_{CD} = 21.2 \text{ kN}$ (tension)
Hence assumption 1 is correct.

Resolve horizontally: $21.2\cos 45 + T_{CE} - 55 = 0$
 $\therefore T_{CE} = 40.0 \text{ kN}$ (tension)

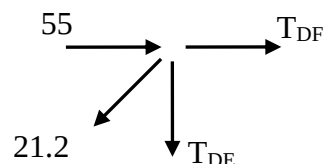


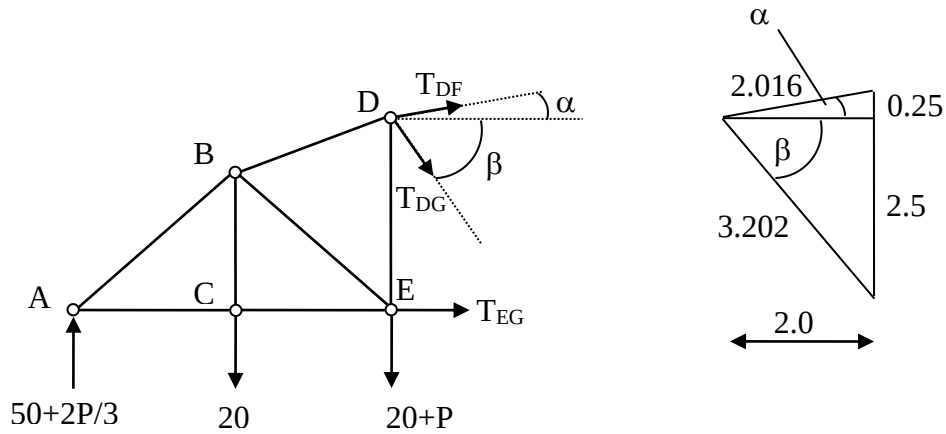
Assumption 2: EF is in tension. Hence, no force exists in DG and this member can be ignored.

Consider joint D

Resolve vertically: $-T_{DE} - 21.2\cos 45 = 0$
 $\therefore T_{DE} = -15.0 \text{ kN}$ (compression)

Resolve horizontally: $T_{DF}\cos 45 - 15 = 0$
 $\therefore T_{DF} = -40.0 \text{ kN}$ (compression)





$$\cos\alpha = 2/2.016 \quad \sin\alpha = 0.25/2.016 \quad \cos\beta = 2.0/3.202 \quad \sin\beta = 2.5/3.202$$

Resolve vertically: $50 + 2P/3 + T_{DF}\sin\alpha - T_{DG}\sin\beta - 20 - (20 + P) = 0$

$$\frac{0.25}{2.016} T_{DF} - \frac{2.5}{3.202} T_{DG} = -10 + \frac{P}{3} \quad [3]$$

Resolve horizontally: $T_{DF}\cos\alpha + T_{DG}\cos\beta + T_{EG} = 0$

$$\frac{2.0}{2.016} T_{DF} + \frac{2.0}{3.202} T_{DG} + T_{EG} = 0 \quad [4]$$

Taking moments about D (+ve clockwise)

$$\left\{ \left(50 + \frac{2P}{3} \right) \times 4 \right\} - (20 \times 2) - (T_{EG} \times 2.5) = 0$$

$$T_{EG} = 64 + \frac{16P}{15} \quad [5]$$

Subs [5] into [4]:

$$\frac{2.0}{2.016} T_{DF} + \frac{2.0}{3.202} T_{DG} = -64 - \frac{16P}{15} \quad [6]$$

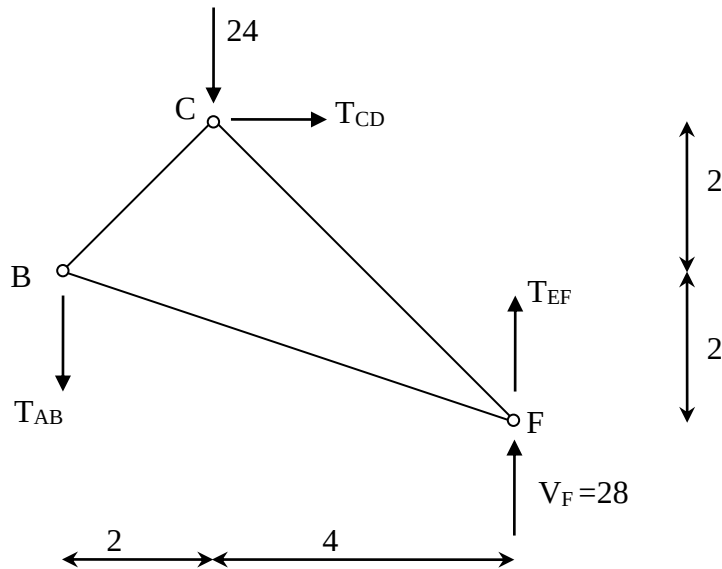
Subtract [6] from 8 x [3]

$$\frac{22}{3.202} T_{DG} = 16 - \frac{56P}{15}$$

$$\text{For } T_{DG} = 0 \Rightarrow 16 - \frac{56P}{15} = 0, \text{ i.e. } P = 16 \times 15 / 56 = \mathbf{4.29 \text{ kN}}$$

- 5 Resolve vertically: $V_A + V_F = 48$ [1]
 Resolve horizontally: $H_A = 12\text{kN}$ (to the left)
 Taking moments about A, (+ve clockwise):
 $(24 \times 2) + (24 \times 4) + (12 \times 2) - (V_F \times 6) = 0 \Rightarrow V_F = 28\text{kN}$ (upwards)
 $\therefore V_A = 20\text{ kN}$ (upwards)

Cut out the triangle BCF from the structure, i.e. **not a straight cut!**



Resolve vertically: $28 + T_{EF} - T_{AB} - 24 = 0$
 $T_{EF} - T_{AB} = -4$

Resolve horizontally: $T_{CD} = 0$

Take moments about point F (+ve clockwise): $4T_{CD} - 6T_{AB} - 24 \times 4 = 0$
 $\therefore T_{AB} = -16\text{ kN}$ (compression)

Hence, $T_{EF} = -20\text{ kN}$ (compression)
